Daniel Mora

New York, NY · Authorized to work in the US · Open to relocation

AI Systems & Research Engineer

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PROFILE

I build ML systems end to end, from point-in-time data and self-supervised pretraining to custom GPU kernels and unattended agentic pipelines. My research uses sparse autoencoders to test whether language models and the brain share features.

EDUCATION

IE University — Madrid, Spain · 2026

Bachelor in Computer Science and Artificial Intelligence

EXPERIENCE

Quantitative Research Intern — WhiteHole — Jun–Nov 2025

Hedge fund; work under NDA, described at a high level

- Built macroeconomic forecasting models (CPI, GDP) from housing-market data.
- Built NLP models linking public political communications to market volatility, including regime analysis.
- Built a knowledge-graph pipeline from NLP-extracted entities and relations using locally hosted language models.

Front-end Developer Intern — MAI-Book — Remote · Jun–Oct 2023

- Built user-facing web interfaces for an online library and ran A/B tests and UX research to improve engagement.

Earlier: IE Start-Up Lab (Jan–Jul 2023)

SELECTED WORK

JEPA Transformer — ArgusJEPA + CassaFormer — PyTorch · Triton · 2026

Two-stage forecaster for US equities; code on GitHub, weights released on Hugging Face

- Point-in-time data platform (IBKR, SEC by acceptance time, FRED/ALFRED vintages, scored news) on PostgreSQL + DuckDB; top-500, survivorship-free.
- Self-supervised JEPA encoder (EMA teacher, 50%-masked views) frozen into 256-d daily embeddings for a separately trained 2.05M-parameter causal decoder with RoPE, QK-RMSNorm, top-1 MoE and a Student-t head.
- Fused Triton kernels, kept only where they beat `torch.compile` on an L4: MoE experts 1.2–4.1× per layer, 1.1–1.5× end to end.

TC_ACC — Autonomous Documentary Studio — Python · LangGraph · Remotion · 2026

Agentic production system; full code private, public excerpt on GitHub

- A case reference in; a researched, narrated, scored and rendered 10-minute documentary out in about five hours, with no human intervention.
- 29 stages, 6 departments, 2 review gates; typed Pydantic artifacts, resumable stages, 46 CI tests pass with provider keys blanked.

Brain–LLM Feature Correspondence — Bachelor thesis — Supervisor: Dr. Luciano Dyballa

Draft paper (ICLR format): “A Cross-Description Test of Sparse Feature Correspondence Between Language Models and fMRI Responses”

- Tested whether brain and LM mappings rely on the same sparse features: 59 participants, 200 cortical parcels, six SAE systems (1B–12B).
- Built SAE feature discovery, encoding models and structured nulls; the calibrated result bounds strong correspondence.

Sparse-autoencoder releases — PyTorch · Hugging Face · GCP

- **OLMo SAE Builder:** streamed ∼20M activations per layer from OLMo-3-7B-Think; released seven SAELens-compatible TopK SAEs.
- **Triplet SAE Project:** SAE steering vs. LoRA for factual triplet extraction; shipped a fine-tuned Gemma 3 4B with GPTQ exports.

LEADERSHIP

- **Founder, IE Quant Finance Club**
- **Founder & President, IE Dominican Republic Club** — grew its flagship event into the university’s largest student-led diversity event (500+ attendees).
- **Business Development Officer, IE Investment Club** — secured sponsorships with financial-sector firms.

SKILLS

ML & research	PyTorch, Triton, Hugging Face (Transformers, PEFT/LoRA, SAELens), SAEs, JEPA, MoE, NLP, forecasting
Systems	Python, C++, SQL, TypeScript; LangGraph, Pydantic, FastAPI; PostgreSQL, DuckDB, Redis; Docker, GitHub Actions, GCP, Azure
Languages	Spanish (native), English (fluent), French (B1)